Train a Human Neural Network KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

The club: Fighters and Wizards can join. Giants and Minis can't. **Strong** = 4 or 5 stars = **1**. Not strong = **0**.

Fire: a neuron stands up if its total is **more than** its threshold.

Fix rule: Should have fired: **ADD** 1 to each weight whose input was 1, threshold −1. Fired but shouldn't: **SUBTRACT**.

Who's to Blame? Find the hidden neuron closest to its threshold. Would the answer be right if it changed? Yes: it fixes itself. No: the output fixes itself.

1. Can one neuron do it? Try weights for one neuron. Test all 4 creatures.

Strong Strength weight _____ Strong Magic weight _____ Threshold _____ Right: _____ of 4

Did anyone get 4 of 4? no Why not? A Giant has both, so its total is even bigger.

2. Pass 1. Start with the numbers on the board. Write the totals for each card.

Card	Inputs	Hidden 1 total	Hidden 2 total	Output total	Says	Right?	Who fixes?
T1 Konik	1, 0	−1	0	0	can't	no	Output
T2 Pasir	0, 1	1	−1	0	can't	no	Hidden 1
T3 Vasi	1, 1	1	−1	1	can join	no	Hidden 1
T4 Skape	0, 0	0	0	0	can't	yes	nobody
T5 Porris	1, 0	−2	0	0	can't	no	Hidden 2
T6 Galla	0, 1	1	−1	0	can't	no	Hidden 1
T7 Gilax	1, 1	0	0	0	can't	yes	nobody
T8 Grapis	0, 0	0	0	0	can't	yes	nobody

Mistakes in pass 1: 5

3. The numbers on the board. After each fix, write all 9 numbers.

		Hidden 1			Hidden 2			Output		
	Card	Strength	Magic	thresh.	Strength	Magic	thresh.	Hidden 1	Hidden 2	thresh.
Start		−1	+1	1	0	−1	1	+1	+1	1
Fix 1	T1	−1	1	1	0	−1	1	1	1	**0**
Fix 2	T2	−1	**2**	**0**	0	−1	1	1	1	0
Fix 3	T3	−2	**1**	**1**	0	−1	1	1	1	0
Fix 4	T5	−2	1	1	**1**	−1	**0**	1	1	0
Fix 5	T6	−2	**2**	**0**	1	−1	0	1	1	0
Fix 6										

Word bank: Fighter Giant Mini Wizard

You can use a word more than once.

4. Pass 2. Same cards, same order. Use your numbers from the end of pass 1.

Card	Hidden 1 total	Hidden 2 total	Output total	Says	Right?
T1 Konik	−2	1	1	can join	yes
T2 Pasir	2	−1	1	can join	yes
T3 Vasi	0	0	0	can't	yes
T4 Skape	0	0	0	can't	yes

Card	Right?
T5 Porris	yes
T6 Galla	yes
T7 Gilax	yes
T8 Grapis	yes

Mistakes in pass 2: 0 (If you made mistakes, do another pass on the back.)

5. What did the hidden neurons learn?

Hidden 1 fires only for a Wizard. Hidden 2 fires only for a Fighter.

Did anyone tell them to look for these? no How did they learn it? By fixing mistakes.

6. Test cards. Do **not** fix anything on a test!

Card	Stars	Inputs	Network says	Really (unfold)
X1 Kromos	5, 2	1, 0	can join	Fighter: can join
X2 Vepul	2, 4	0, 1	can join	Wizard: can join
X3 Tarra	5, 4	1, 1	can't	Giant: can't
X4 Volar	1, 2	0, 0	can't	Mini: can't
X5 Tibi	3, 2	0, 0	can't	Fighter: can join
X6 Gobur	3, 5	0, 1	can join	Giant: can't

Why did the network get Tibi wrong? Its Strength is only 3 stars, so it looks like a Mini.

7. Talk about it. One neuron couldn't learn the club rule because a Giant has both.

Go Further (Day 2). Same cards, different start. Use the tool on the lesson page.

Start B: did it learn? yes Hidden 2 fires only for a Giant. Start C: did it learn? no

What happened with Start C? It got stuck. It went around in circles.